Group Assignment 3: A Bayesian Subgroup Comparison of the French Short Boredom Proneness Scale

Nikola Sekulovski & Maarten Marsman

Information about the assignment [not finished]

In this group assignment, you will **reanalyze an existing dataset** using the Bayesian graphical modeling methodology for group comparisons that you just learned. You are provided with a short study description and a **specified Methods section**. Your task is to:

- Complete the definition of the inclusion Bayes factor in the Methods section
- Implement the analysis exactly as implied by the Methods section
- Write a very brief **Results section** where the results are summarized.

You will work with the **French subsample (N = 490)** from the original study, using **18 ordinal items** from the **Short Boredom Proneness Scale (SBPS)** and the **Curiosity and Exploration Inventory–II (CEI-II)**.

Introduction

Boredom proneness has been conceptualized as a trait-like tendency to experience boredom frequently and intensely, and has been linked to a broad range of psychological outcomes. The Short Boredom Proneness Scale (SBPS) was developed as a concise unidimensional measure of boredom proneness and has since been translated into multiple languages. In the original validation of the French version of the SBPS, Martarelli et al. (2022) examined the relationships between boredom proneness (SBPS), and measures on two subscales of the Curiosity and Exploration Inventory (CEI-II), using a regularized frequentist network approach. Their analysis suggested positive associations between boredom and curiosity items and negative associations between boredom and exploration items. However, the authors did not investigate whether the estimated network structure differs between female and male participants.

The aim of the present study is to reanalyze the French subsample of the original dataset using Bayesian graphical modeling with the specific focus on testing whether the network structure differs between female and male participants.

Method

Data

The data consists of responses from N = 490 French-speaking participants ($M_{age} = 33.8$; 73% females and 27% males). The network includes 18 items, all measured on a 7-point Likert scale:

- 8 items from the Short Boredom Proneness Scale (SBPS)
- 10 items from the Curiosity and Exploration Inventory–II (CEI-II)

Missing data was handled by listwise deletion.

Model Specification

To model the differences the differences in the network structure we used ….. (Marsman et al., in press).

Prior Distributions

Structure Priors

For the inclusion of edge differences we assigned independent Bernoulli priors with a default inclusion probability of 0.5.

Parameter Priors

The difference parameters were assigned independent Cauchy priors with a scale of 2.5.

Inference

Inference (this is adjusted text from the morning session assignment)

Evidence for the presence or absence of differences was quantified using the inclusion Bays factor. The inclusion Bayes factor ____. These Bayes factors were used to classify edges as showing evidence for _________, evidence for _________, or _________ evidence.

_________ plots were used to visualize which edges fall into each of these categories. In this study, an inclusion Bayes factor of _________ was used as the threshold for strong evidence for inclusion, and a Bayes factor of _________ was used as the threshold for strong evidence for exclusion; values falling between these thresholds were classified as inconclusive.

Posterior summaries of edge parameter estimates for the differences were obtained from the model-averaged posterior distribution from which posterior mean and HDI intervals were calculated. Centrality measures were not estimated or interpreted.

Software

We used the easybgm R package (Huth et al., 2024) to estimate the the model and plot the results. We ran the model using an adaptive Metropolis–Hastings algorithm, with all other sampling settings left at their default values.

Results

Evidence for Edge Differences

Parameter Estimates

References

- Huth, K., Keetelaar, S., Sekulovski, N., van den Bergh, D., & Marsman, M. (2024). Simplifying Bayesian analysis of graphical models for the social sciences with easybgm: A user-friendly R-package. *advances.in/psychology, e66366* doi: 10.56296/aip00010

- Martarelli, C. S., Baillifard, A., & Audrin, C. (2022). A trait-based network perspective on the validation of the french short boredom proneness scale. *European Journal of Psychological Assessment, 39,* 390–399 doi: 10.1027/1015-5759/a000718346
- Marsman M, Waldorp LJ, Sekulovski N, Haslbeck JMB (in press). “Bayes factor tests for group differences in ordinal and binary graphical models.” *Psychometrika.* doi: 10.1017/psy.2025.10060